



PROJECT STATUS REPORT · MAY 2026

FG Inventory Management System

AI-Powered Food & Grains ERP — Build Progress & Technical Overview

COORDINATOR	PHASE	DATE	STATUS	CLASSIFICATION
Prachi	3 of 4	25 May 2026	Active Build	Confidential

SECTION 01

Project Overview

FG Inventory is a production-grade, web-based ERP system built for a food & grains wholesale business. It replaces manual stock registers with a real-time intelligence platform — barcode-driven flows, FIFO expiry tracking, AI-generated operational insights, and an EPR compliance module.

TECHNOLOGY STACK

FRONTEND

React 18Vite 5

Recharts

React Router v6

Tailwind v4 (EPR)

BACKEND

Node.jsExpress

Knex.jsJWT Auth

SheetJSnode-cron

DATABASE

PostgreSQL

Neon (Cloud)

FIFO Locking

Advisory Locks

INFRA & AUTH

JWT + RBAC5 Roles

Axios Interceptors

Rate Limiting

USER ROLES & ACCESS

ROLE	PERMISSIONS	KEY PAGES
Admin	Full access + user management + all reports	All pages, Users, Audit Log, System Health, Normalize
Purchase	Inward entry, PO creation, vendor management	Purchase Orders, Inward, Vendors
Warehouse	Receive goods, barcode scan, stock view	Scan, Items, Batches, Stock Transfers
Sales	Outward dispatch, challan generation	Outward, Customers, Challan
View-only	Read-only access to reports	Dashboard, Reports, MIS, Expiry Alerts

CORE BUSINESS MODULES



INVENTORY ENGINE

FIFO batch tracking · Expiry management · Dynamic reorder points · Live stock valuation



TRANSACTION FLOWS

PO → Inward → Confirm → Lock · Dispatch → FIFO pick → Challan → Lock



AI INTELLIGENCE

Velocity scoring · Demand trends · Stockout risk · Auto-generated operational insights







EPR COMPLIANCE

PWM Rules 2022 · 8 material classes · Liability computation · Recyclability matrix

ALERTS & RISK

Low-stock alerts · Expiry risk badges · Nightly batch risk scoring (0–100)

ANALYTICS & MIS

Margin tracker · Shrinkage cost · Dead stock detection · Monthly MIS dashboard

SECTION 02

Before the Build

The state of inventory management before FG Inventory was implemented — manual processes, fragmented records, zero visibility into stock health or compliance obligations.

STOCK MANAGEMENT

Manual register entries · No batch tracking · FIFO not enforced · Stock counts done weekly by hand · Expiry dates tracked on paper · No reorder triggers

INWARD / OUTWARD

Excel sheets per vendor · No PO linkage · Manual challan writing · No confirmation or lock process · Corrections by erasing entries · Zero audit trail

REPORTING & VISIBILITY

No real-time dashboard · Month-end reconciliation only · Margin calculated manually in Excel · No dead stock identification · No vendor performance data

COMPLIANCE & ACCESS

No user roles or access control · Anyone could edit any record · EPR obligations unknown · No audit log · No barcode scanning — all entry manual

KEY PAIN POINTS VS. IMPACT

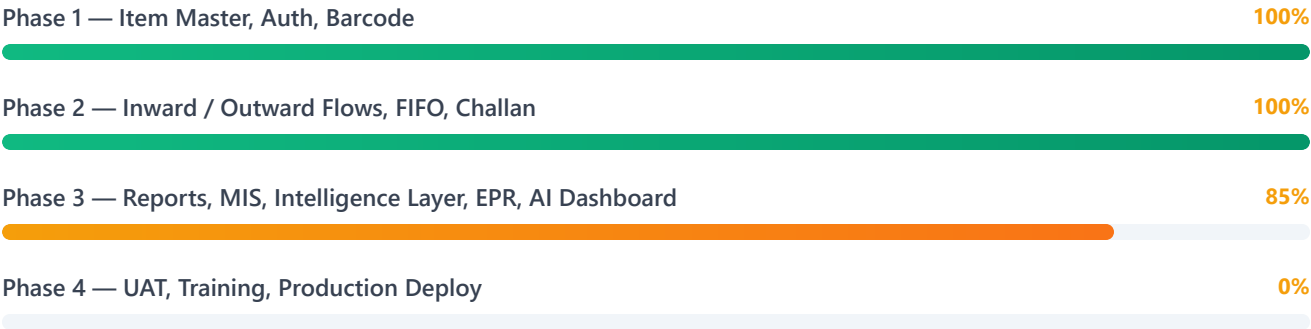
PAIN POINT	ROOT CAUSE	BUSINESS IMPACT	SEVERITY
Stock spoilage	No FIFO enforcement, expiry not tracked per batch	Direct write-off loss on expired goods	CRITICAL
Stockouts	No reorder triggers; stock counted weekly	Lost sales, emergency purchases at high cost	CRITICAL
Reconciliation errors	Manual challan ↔ invoice ↔ register mismatch	Revenue leakage, vendor disputes	CRITICAL
No audit trail	Single Excel file, anyone edits	Fraud risk, no accountability	CRITICAL
EPR non-compliance	PWM Rules 2022 obligations unknown	Regulatory penalty risk	HIGH
No margin visibility	Dispatch rate vs purchase rate never compared	Pricing decisions made blind	HIGH
No barcode scanning	All entry manual, high error rate	Slow operations, data inaccuracy	HIGH

SECTION 03

After Build — Current Status

What has been delivered, what is live, and what remains. Phase 3 of 4 — entering UAT preparation.

PHASE PROGRESS



BEFORE VS. AFTER COMPARISON

✕ BEFORE	FEATURE AREA	✓ AFTER (LIVE)
Paper register, no batch IDs	Stock Tracking	Live batch ledger, FIFO enforced, qty_remaining per batch in real-time
Manual challan writing	Outward Dispatch	FIFO auto-pick → Challan auto-generated → Lock with audit trail
Excel sheets, no PO workflow	Inward / Purchasing	PO → Inward → Confirm → Lock → Batch created automatically
Weekly manual count	Stock Alerts	Dynamic ROP — alerts fire when live stock ≤ reorder point (nightly recalc)
No expiry awareness	Expiry Management	Per-batch expiry · Risk score 0–100 · Expired qty auto-zeroed nightly
No barcode system	Barcode Scanning	EAN-13 auto-generated · HID scanner input · Scan → item lookup in <1s
No visibility, month-end only	Dashboard & Reporting	AI Command Center — velocity leaders, demand trends, insights, live activity
EPR obligations unknown	EPR Compliance	Full EPR dashboard — 8 material classes, liability calc, recyclability matrix
No access control	Security & Roles	JWT + 5 RBAC roles · Rate-limited login · Audit log for every write action

✕ BEFORE	FEATURE AREA	✓ AFTER (LIVE)
No margin data	Margin & MIS	Per-item margin %, shrinkage cost, dead stock detection, Margin Report page

SECTION 04

Feature Delivery Log

Full timeline of what was built, what's in active development, and what's queued next.

✓

Phase 1 — Foundation

COMPLETE

Item Master CRUD · Auto item_code (FG-0001) · EAN-13 barcode generation · Category + Sub-category structure · User auth (JWT) · 5-role RBAC middleware · Login page · Barcode scanner input · Label print

✓

Phase 2 — Transaction Flows

COMPLETE

Purchase Orders · Inward (PO→Receive→Scan→Confirm→Lock) · Outward (Dispatch→FIFO pick→Challan→Lock) · Batch creation on inward confirm · FIFO with row-level locking (.forUpdate()) · Challan auto-number with advisory lock · Immutable locked records · Reversal entry pattern

✓

Phase 3A — Intelligence & Analytics

COMPLETE

Nightly cron: risk scoring (0–100) · Dynamic ROP recalc · Expired batch auto-zero · Dead stock detection query · Margin & shrinkage per item · MIS Dashboard API · Vendor history · Excel export (SheetJS) · Expiry Alerts page · Margin Report page · Batch Viewer

✓

Phase 3B — AI Command Center

COMPLETE

New /api/intelligence/command-center endpoint · Velocity scoring (kg/day 30-day rolling) · Demand trend (last 7d vs prior 7d) · Stockout risk levels (critical/high/medium/low) · Auto-generated AI insights (no external API) · Dashboard rebuilt — dark hero header, 7 KPI strip, velocity leaders table, Recharts category chart, live activity feed, insights panel

✓

Phase 3C — EPR Compliance Dashboard

COMPLETE

EPR module (India PWM Rules 2022) · 8 material classes (HDPE, LDPE, MLP, PET-01, ALU-CAN, Liquid Carton, Paperboard) · Liability computation engine · Recyclability color matrix (●●●) · DonutChart (pure SVG) · 5 retail categories × 24 sub-categories · Full PostgreSQL schema + seed data

✓

Phase 3D — Security Hardening

COMPLETE

FIFO race condition fixed (forUpdate row lock) · Challan sequence serialised (pg_advisory_xact_lock) · 401 redirect loop eliminated (_redirecting flag) · Login brute-force protection (5 attempts / 15 min per IP) · Direct fetch() calls replaced with axios client · Bulk PATCH wrapped in transaction · Query injection guards on filter params

✓

Phase 3E — Food Photography System

COMPLETE

Wikipedia REST API thumbnail pipeline (no API key, CORS-enabled) · 35+ food categories mapped to correct Wikipedia articles · Async load with emoji placeholder → real photo crossfade · Session-level cache + concurrent fetch deduplication · prewarmImages() fires on Items page load · Custom item_image_url always takes priority

🔄

In Progress

ACTIVE

Items page grid/list view toggle · Thumbnail-first InventoryGridCard component · Velocity badges on Items page (🔥 Fast / ⚡ Active / 🐢 Slow) · Stock health indicator bars

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Phase 4 — UAT & Production

QUEUED

User acceptance testing with Prachi · Staff training sessions · Production deployment · Mobile sidebar collapse · Push notifications for low stock · Pet Pooja API integration (optional)

SECTION 05

Developer Skills Matrix

Required technical competencies to continue building and maintaining FG Inventory through Phase 4.

SKILL	LEVEL	HOW IT APPLIES TO THIS PROJECT
React 18 + Hooks	Expert	Component architecture, async state, FoodPhoto crossfade, velocity leaders table, AI insight cards
Node.js + Express	Expert	25+ REST routes, middleware chains, role guards, error handling, streaming blob downloads
PostgreSQL + Knex.js	Expert	FIFO batch queries, window functions, .forUpdate() row locks, pg_advisory_xact_lock, CTE velocity SQL
Business Logic Design	Expert	FIFO dispatch, ROP formula, risk scoring (0–100), margin/shrinkage model, EPR liability calc
JWT + RBAC Security	Advanced	5-role system, route + UI gating, brute-force rate limiter, 401 redirect de-duplication
Data Visualisation	Advanced	Recharts BarChart / ResponsiveContainer, pure SVG DonutChart (EPR), stock health progress bars
Async / Performance Patterns	Advanced	Promise.allSettled for parallel API calls, Wikipedia fetch deduplication, session-level image cache, prewarm
CSS / Design Systems	Advanced	All UI via JS style objects + CSS custom properties (no className) · Dark hero headers · Glassmorphism patterns
Excel / SheetJS	Intermediate	Inward template download, import row validation, export endpoints for all report types
Barcode Systems	Intermediate	EAN-13 generation, HID scanner keyboard emulation, focused input capture, scan-to-lookup flow
Cron / Scheduled Jobs	Intermediate	Nightly 2am job: risk scoring, ROP recalc, batch expiry zeroing, cron_log table
Tailwind CSS v4	Intermediate	EPR dashboard only — PostCSS plugin setup, utility-only import avoiding preflight conflict with existing app
Regulatory / Compliance	Intermediate	India EPR PWM Rules 2022 — material class rates, recyclability grading, waste log schema

PROJECT BY THE NUMBERS

